Master's Degree in Data Science
Academic Year: 2025-2026

HOMEWORK 1

CUSTOMER CHURN PREDICTION

MARKETING ANALYTICS

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DATASETS & PREPROCESSING

Dataset	Samples	Target Distribution
Dataset 1	40,000	52% churn - Balanced
Dataset 2	50,000	87% churn - Imbalanced

DATASET OVERVIEW & BUSINESS SCENARIOS

The analysis evaluates two distinct market scenarios across datasets sharing identical structures (13 features, 1 target variable):

- **Dataset 1** represents a **baseline scenario** in which roughly 1 out of 2 customers churns.
- **Dataset 2** captures a **high-risk business scenario**, where the majority of the customer base leaves.

TRAIN-TEST SPLIT

Datasets are split into training (75%) and test (25%) sets to ensure the evaluation of the model capability on unseen data. In particular, **stratified random sampling**, which preserves the exact churn/no-churn ratio across both training and test, is fundamental especially for the heavily skewed Dataset 2.

DATA PREPROCESSING

- **Imputation** steps are included into the pipeline for robustness, but both datasets do not have missing values.
- **Standard scaling** is applied to numeric variables to normalize distributions.
- **Dummy encoding** is applied to categorical variables to obtain model-readable binary vectors.

Features
Numeric: age, income, tenure_months, monthly_spend, website_visits, support_tickets, email_click_rate, loyalty_score, discount_use, last_campaign_days
Categorical: gender, region, campaign_type

MODELS & HYPERPARAMETERS

Model	Hyperparameter	Values
Logistic Regression	c	[0.01, 0.1, 0.5, 1.0]
Random Forest	n_estimators	[50, 100, 150]
	max_depth	[5, 10, 15]
Gradient Boosting	n_estimators	[50, 100, 150]
	learning_rate	[0.05, 0.1, 0.2]

MODEL SELECTION

The **F1 score** is adopted as the primary measure for model selection. By balancing precision and recall, it addresses a critical trade-off between maximizing budget efficiency and capturing churning customers. In particular:

- Dataset 1 - **Logistic Regression**: in a balanced market environment, a linear boundary proves highly effective, ensuring model interpretability and parsimony.
- Dataset 2 - **Gradient Boosting**: as a sequential boosting ensemble, it excels at capturing complex, non-linear patterns, maximizing predictive power in the context of a highly skewed class distribution.

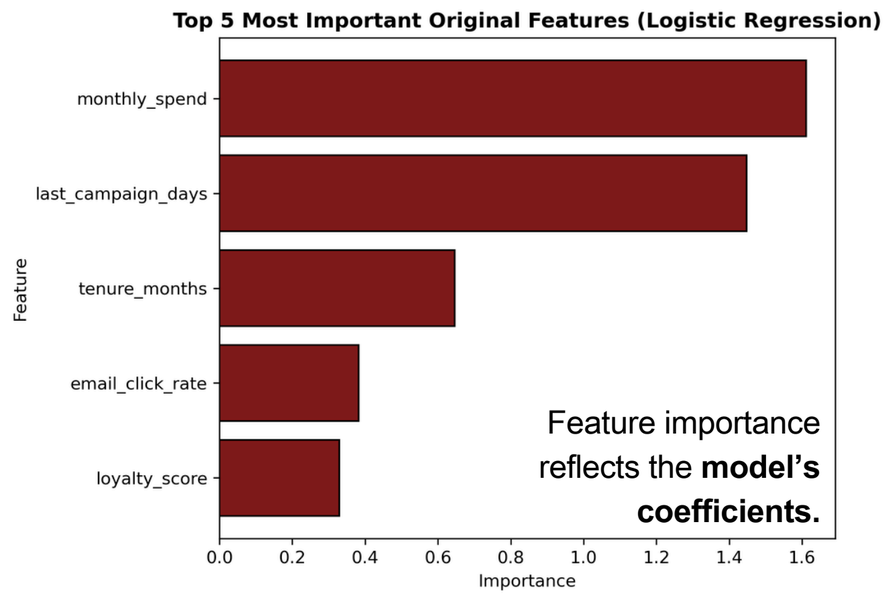
MODEL TRAINING & TUNING

Three machine learning models are selected: Logistic Regression, Random Forest, and Gradient Boosting. In particular, **hyperparameter tuning** is conducted using the **validation set** extracted from the training set to avoid data leakage. Once the optimal configurations are identified, models are **retrained on the full training set** and **evaluated on the test set**.

Dataset	Best Model	Hyperparameter	Value
Dataset 1	Logistic Regression	c	0.01
Dataset 2	Gradient Boosting	n_estimators	100
		learning_rate	0.2

EVALUATION & FEATURE IMPORTANCE

DATASET 1 (LOGISTIC REGRESSION)

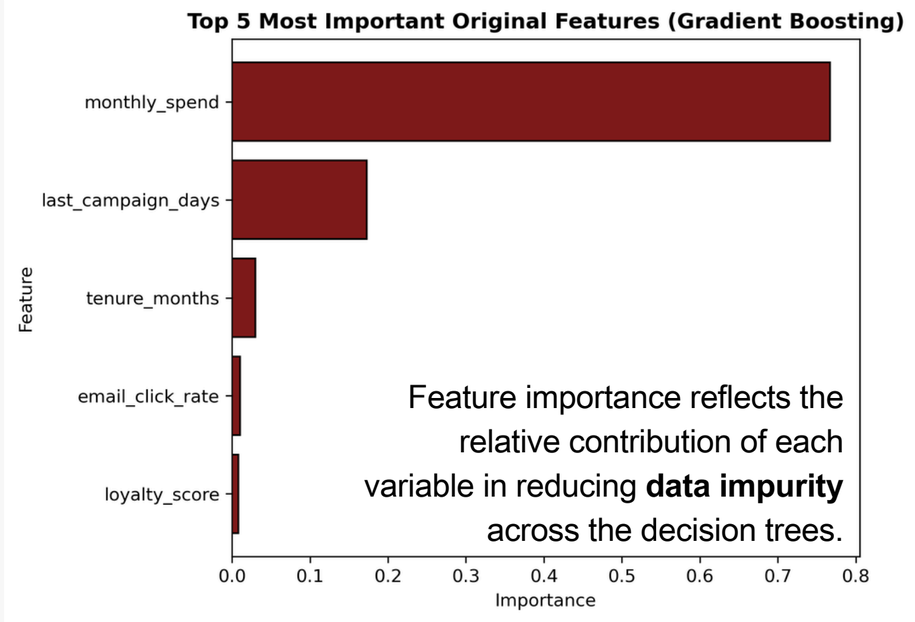


Confusion Matrix		Predicted	
		0	1
Actual	0 (no churn)	3,931	891
	1 (churn)	927	4,252

Metric	Value
Accuracy	0.82
F1 score	0.82
ROC-AUC	0.90

Logistic Regression shows balanced performance, being equally effective at predicting both churners and non-churners.

DATASET 2 (GRADIENT BOOSTING)



Confusion Matrix		Predicted	
		0	1
Actual	0 (no churn)	1,125	466
	1 (churn)	270	10,640

Metric	Value
Accuracy	0.94
F1 score	0.97
ROC-AUC	0.97

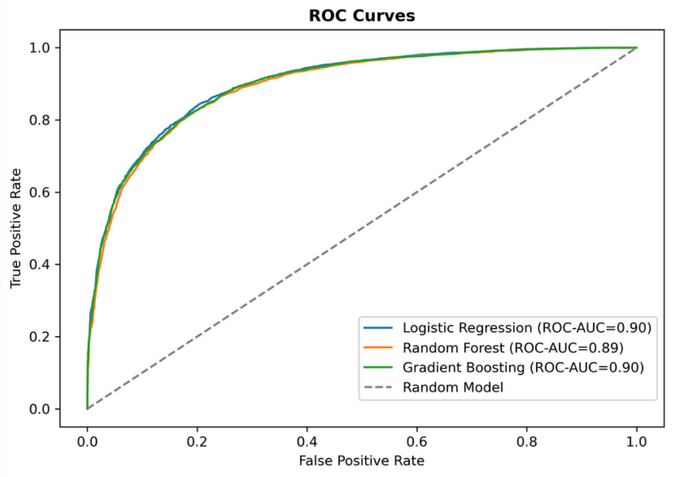
Gradient Boosting achieves high overall metrics, driven by the dominant class, but it misclassifies nearly 29% of the rare non-churning minority.

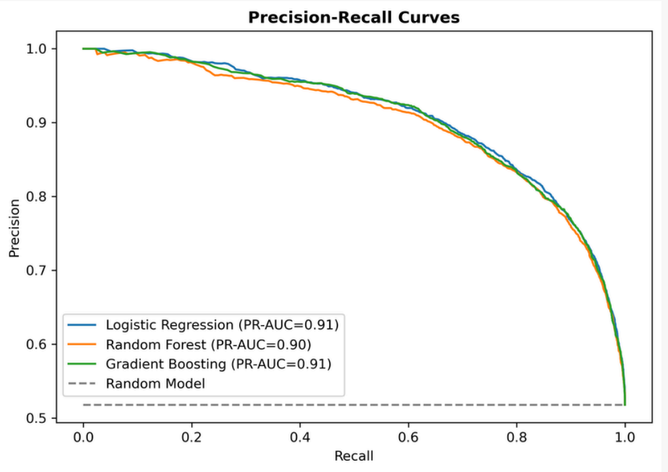
MULTI MODELS COMPARISON

Model comparison is based on evaluating the top-performing configuration of each of the three models for both datasets.

DATASET 1

	Linear Regression	Random Forest	Gradient Boosting
Accuracy	0.82	0.81	0.81
F1 score	0.82	0.82	0.82
ROC-AUC	0.90	0.89	0.90





DATASET 2

	Linear Regression	Random Forest	Gradient Boosting
Accuracy	0.90	0.94	0.94
F1 score	0.94	0.96	0.97
ROC-AUC	0.97	0.96	0.97

